

Unit ID: 6798d169aaf593b2823f1828

Unit 5 - Knowledge Check

PDF Documents

PDF Document 1: 1738068327695-Chapter 7_unit 5 - Knowledge Check.pdf

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Update #1 - 2025-05-16 15:19:25

KNOWLEDGE CHECK

1 What is a ratio?

- ☐ A A comparison of two numbers by division.
- ☐ B A type of mathematical operation.
- ☐ C A formula for finding the perimeter.
- ☐ D A method to solve equations.

CORRECT ANSWER

☐ A

2 If the ratio of water to cement in a mixture is 3:1, how many units of water are needed for 4 units of cement?

- ☐ A 1 unit
- ☐ B 2 units
- ☐ C 4 units
- ☐ D 12 units

CORRECT ANSWER

☐ D

3 Which scenario describes a use of ratios in cooking?

- ☐ A Measuring a cup of flour.
- ☐ B Mixing 2 parts oil to 1 part vinegar for salad dressing.
- ☐ C Boiling water for 10 minutes.
- ☐ D Slicing a pie into 8 pieces.

CORRECT ANSWER

☐ B

4 In a ratio 5:3, what does '5' represent if the ratio is describing the number of apples to oranges?

- ☐ A The total fruits
- ☐ B The number of oranges
- ☐ C The number of apples
- ☐ D Neither apples nor oranges

CORRECT ANSWER

☐ C

5 What is a variable in programming?

- ☐ A A place to store music files.
- ☐ B A type of programming language.
- ☐ C A storage location named for a type of data that can change during program execution.
- ☐ D A fixed number used in calculations.

CORRECT ANSWER

☐ C

6 Which logical operator would be used to check if two conditions are both true?

- ☐ A AND
- ☐ B OR
- ☐ C NOT
- ☐ D XOR

CORRECT ANSWER

☐ A

7 In programming a timer, what does a 'nested-if' structure allow the programmer to do?

- ☐ A Run the same code multiple times.
- ☐ B Check for multiple conditions sequentially within a block of code.
- ☐ C Increase the speed of the program.
- ☐ D Connect to the internet.

CORRECT ANSWER

☐ B

Critical Thinking Challenge

- 1** How could understanding ratios help you when shopping for ingredients to make a meal for a large group of people? Discuss how you would use ratios to scale up a recipe.

- 2** Consider a situation where you and your friends are sharing the cost of a gift. If you use a ratio to determine how much each person should contribute based on their income, what factors would you consider to ensure fairness?